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Activity 3: Install SSH server on CentOS or RHEL 8	
1. Objectives: 1.1 Install Community Enterprise OS or Red Hat Linux OS 1.2 Configure remote SSH connection from remote computer to CentOS/RHEL-8	
2. Discussion: CentOS vs. Debian: Overview CentOS and Debian are Linux distributions that spawn from opposite ends of the candle. CentOS is a free downstream rebuild of the commercial Red Hat Enterprise Linux distribution where, in contrast, Debian is the free upstream distribution that is the base for other distributions, including the Ubuntu Linux distribution. As with many Linux distributions, CentOS and Debian are generally more alike than different; it isn't until we dig a little deeper that we find where they branch. CentOS vs. Debian: Architecture The available supported architectures can be the determining factor as to whether a distro is a viable option or not. Debian and CentOS are both very popular for x86_64/AMD64, but what other archs are supported by each? Both Debian and CentOS support AArch64/ARM64, armhf/armhfp , i386 , ppc64el/ppc64le. (Note: armhf/armhfp and i386 are supported in CentOS 7 only.) CentOS 7 additionally supports POWER9 while Debian and CentOS 8 do not. CentOS 7 focuses on the x86_64/AMD64 architecture with the other archs released through the AltArch SIG (Alternate Architecture Special Interest Group) with CentOS 8 supporting x86_64/AMD64, AArch64 and ppc64le equally. Debian supports MIPSel, MIPS64el and s390x while CentOS does not. Much like CentOS 8, Debian does not favor one arch over another —all supported architectures are supported equally. CentOS vs. Debian: Package Management Most Linux distributions have some form of package manager nowadays, with some more complex and feature-rich than others. CentOS uses the RPM package format and YUM/DNF as the package manager. Debian uses the DEB package format and dpkg/APT as the package manager.	

Both offer full-feature package management with network-based repository support, dependency checking and resolution, etc.. If you're familiar with one but not the other, you may have a little trouble switching over, but they're not overwhelmingly different. They both have similar features, just available through a different interface.

Task 1: Download the CentOS or RHEL-8 image (Create screenshots of the following)

1. Download the image of the CentOS here:
http://mirror.rise.ph/centos/7.9.2009/isos/x86_64/
2. Create a VM machine with 2 Gb RAM and 20 Gb HD.
3. Install the downloaded image.
4. Show evidence that the OS was installed already.

Task 2: Install the SSH server package *openssh*

1. Install the ssh server package *openssh* by using the *dnf* command:
\$ dnf install openssh-server
2. Start the *sshd* daemon and set to start after reboot:
\$ systemctl start sshd
\$ systemctl enable sshd
3. Confirm that the sshd daemon is up and running:
\$ systemctl status sshd

```
[benedicto@vbox ~]$ systemctl start sshd
[benedicto@vbox ~]$ systemctl enable sshd
[benedicto@vbox ~]$ systemctl status sshd
● sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-02-19 17:07:26 PST; 9min ago
     Docs: man:sshd(8)
           man:sshd_config(5)
  Main PID: 1030 (sshd)
    Tasks: 1 (limit: 61025)
   Memory: 2.6M (peak: 2.8M)
      CPU: 11ms
   CGroup: /system.slice/sshd.service
           └─1030 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Feb 19 17:07:26 localhost.localdomain systemd[1]: Starting OpenSSH server daemon...
Feb 19 17:07:26 vbox sshd[1030]: Server listening on 0.0.0.0 port 22.
Feb 19 17:07:26 vbox sshd[1030]: Server listening on :: port 22.
Feb 19 17:07:26 vbox systemd[1]: Started OpenSSH server daemon.
[benedicto@vbox ~]$
```

4. Open the SSH port 22 to allow incoming traffic:
\$ firewall-cmd --zone=public --permanent --add-service=ssh
\$ firewall-cmd --reload

```
[benedicto@vbox ~]$ sudo firewall-cmd --permanent --add-service=ssh
Warning: ALREADY_ENABLED: ssh
success
[benedicto@vbox ~]$ firewall-cmd --reload
success
```

5. Locate the ssh server man config file `/etc/ssh/sshd_config` and perform custom configuration. Every time you make any change to the `/etc/ssh/sshd-config` configuration file reload the `sshd` service to apply changes:

\$ systemctl reload sshd

```
[benedicto@vbox ~]$ systemctl reload sshd
```

Task 3: Copy the Public Key to CentOS

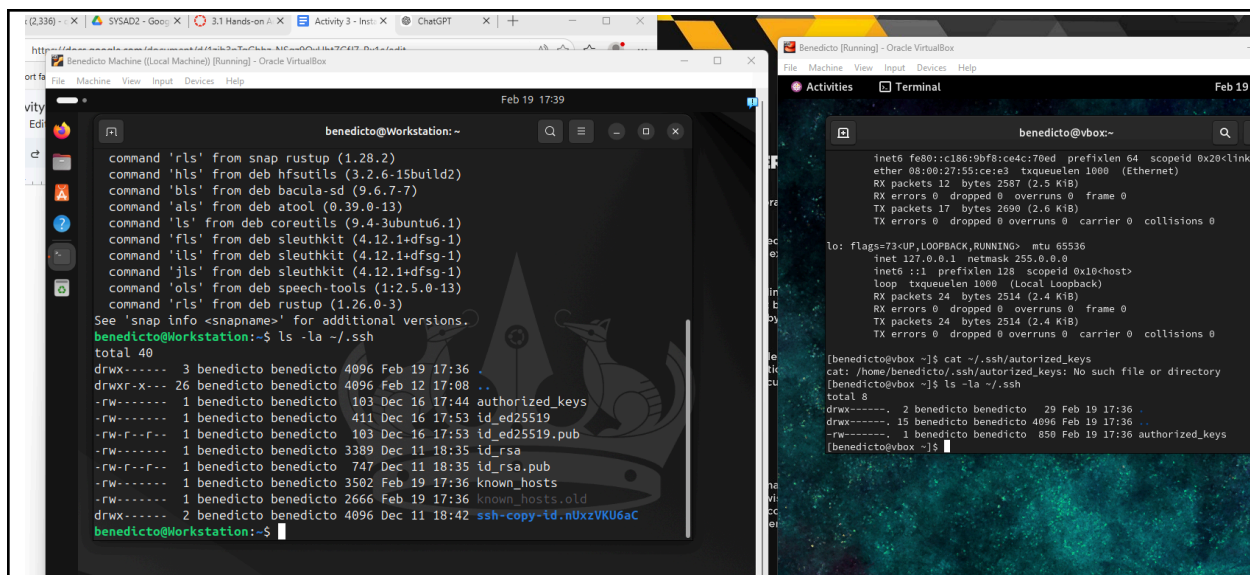
1. Make sure that `ssh` is installed on the local machine.
2. Using the command `ssh-copy-id`, connect your local machine to CentOS.
3. On CentOS, verify that you have the `authorized_keys`.

```
[benedicto@vbox ~]$ sshd --version
unknown option -- -
OpenSSH_9.9p1, OpenSSL 3.5.1 1 Jul 2025
usage: sshd [-46DdeGiqTtV] [-C connection_spec] [-c host_cert_file]
           [-E log_file] [-f config_file] [-g login_grace_time]
           [-h host_key_file] [-o option] [-p port] [-u len]
```

```
benedicto@Workstation:~$ ssh-copy-id benedicto@192.168.56.113
The authenticity of host '192.168.56.113 (192.168.56.113)' can't be established.
ED25519 key fingerprint is SHA256:QCJWUi3onsE9cz5VHPPRdgp2D9u2RoeVs7Ux2/6YMWs.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 2 key(s) remain to be installed -- if you are prompt
ed now it is to install the new keys
benedicto@192.168.56.113's password:

Number of key(s) added: 2

Now try logging into the machine, with:  "ssh 'benedicto@192.168.56.113'"
and check to make sure that only the key(s) you wanted were added.
```



Task 4: Verify ssh remote connection

1. Using your local machine, connect to CentOS using ssh.
2. Show evidence that you are connected.

```
benedicto@Workstation:~$ ssh benedicto@192.168.56.113
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Thu Feb 19 17:31:51 2026
[benedicto@vbox ~]$ 123
bash: 123: command not found...
[benedicto@vbox ~]$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos
[benedicto@vbox ~]$ cd Documents
[benedicto@vbox Documents]$
```

Reflections:

Answer the following:

1. What do you think we should look for in choosing the best distribution between Debian and Red Hat Linux distributions?

- When choosing between Debian and Red Hat Enterprise Linux I think about our purpose, budget, and future goals. If we need a free operating system for school projects, personal use, or learning Linux, Debian is a good choice because it is completely free and supported by a strong community. However, if we plan to work in a business environment or need professional technical support, Red Hat is better because it offers official support and is widely used in companies. We should also consider whether we prefer community help or paid customer service, and whether we want to focus on learning open-source systems or preparing for enterprise jobs.

2. What are the main differences between Debian and Red Hat Linux distributions?

- The main difference between Debian and Red Hat is that Debian is free and community-supported, while Red Hat usually requires a paid subscription and provides official company support. Debian uses the APT package manager and .deb files, while Red Hat uses YUM or DNF and .rpm files. Debian is popular among students, developers, and home users, while Red Hat is mainly used by businesses and large organizations. Both systems are stable and reliable, but Debian focuses more on free and open-source software, while Red Hat focuses on enterprise-level performance and support.